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# Digital Entrepreneurship Policy in Korea, Japan, China, the United Kingdom, the United States, France and Germany, 2022

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Governments of the Republic of Korea, Japan, the People's Republic of China, the United Kingdom, the United States, France and Germany, with their legislatures, the Central Committee of the Communist Party of China and the Federal Cartel Office

*Summary:* This report reviews the digital entrepreneurship policy that the governments of Korea, Japan, China, the United Kingdom, the United States, France and Germany set out in 2022 in six domains and analyses it in its context, against theory and in practice. Data, payments and platform power organised the year's binding measures: China's rules on data exports, Japan's stablecoin law, France's electronic invoicing law, to be phased in, and Germany's special abuse control of large platforms, while AI was governed by principles and voluntary frameworks. The best measures reached digital start-ups directly, as France's DeepNum20 programme did, or checked the platforms they depend on, as Germany's section 19a decisions did; the weakest were the United States' proclamations on culture and China's literacy aims for skills. Much of the year's policy was still a plan, bill or draft at its end; the United Kingdom will close its software voucher scheme and cut SME R&D relief, though it kept its rise in seed relief; and governments reported mostly outputs and claimed effects, with few evaluations.

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EEPR 2022

# Digital Entrepreneurship Policy in Korea, Japan, China, the United Kingdom, the United States, France and Germany, 2022

IFEED Institute for Entrepreneurial Ecosystem Development

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# Digital Entrepreneurship Policy in Korea, Japan, China, the United Kingdom, the United States, France and Germany, 2022

**Abstract**

This report reviews and analyses the digital entrepreneurship policy of the governments of the Republic of Korea, Japan, the People's Republic of China, the United Kingdom, the United States, France and Germany in their own plans, laws, budgets and announcements. The policies are reviewed in six domains — regulation and law, finance, culture, support, human capital and markets — and analysed in their context, against theories of digital entrepreneurship, platforms, regulation, entrepreneurial finance and policy design, and in practice. Data, payments and platforms organised the year's binding measures. China's assessment of outbound data took effect on 1 September and its deep synthesis rules will apply from 10 January 2023; Japan enacted a stablecoin law; France enacted an electronic invoicing duty, to be phased in from a later date; and Germany's Federal Cartel Office placed Alphabet, Meta and Amazon under special abuse control, while platform rules were a draft bill in the United Kingdom and under study in Japan. AI was governed by principles in Japan and the United Kingdom and by a non-binding blueprint and a draft framework in the United States. Support for SMEs' digital adoption ran in China, Korea, France, Germany and the United Kingdom, and France's DeepNum20 programme reached deep-tech digital start-ups, but the United Kingdom will close Help to Grow: Digital. The United States widened practical training for foreign graduates in cloud and data fields, the United Kingdom opened routes for graduates and scale-ups, and routes for founders were plans in Korea and Japan. The United Kingdom will cut its SME R&D relief but extend relief to data and cloud costs and keep larger seed relief, and governments reported mostly outputs and claimed effects, with few evaluations. The period covered is January to December 2022.

**Keywords**

data governance, outbound data transfers, AI governance, stablecoins, crypto-assets, platform competition, regulatory sandboxes, SME digitalisation, electronic invoicing, digital skills, skilled immigration

**JEL Classification**

L26, L86, O33, K24, L51, O38

Entrepreneurial Support Research Division, Institute for Entrepreneurial Ecosystem Development (IFEED).

# 1 INTRODUCTION

**Key findings**

- Digital entrepreneurship policy in 2022 was mostly strategies, plans and proposals; binding measures came on data, payments, invoicing and platform power.
- China's assessment of outbound data took effect on 1 September and its deep synthesis rules will apply from 10 January 2023; Japan enacted a stablecoin law, and AI was governed by principles and voluntary frameworks in Japan, the United Kingdom and the United States.
- Germany's Federal Cartel Office placed Alphabet, Meta and Amazon under special abuse control; platform rules were a draft bill in the United Kingdom and under study in Japan.
- Support for SMEs' digital adoption ran in China, Korea, France, Germany and the United Kingdom, and France's DeepNum20 programme reached deep-tech digital start-ups, but the United Kingdom will close Help to Grow: Digital.
- The United States widened practical training for foreign graduates in cloud and data fields and enacted cyber training for small-business advisers, still to be set up; the United Kingdom opened routes for graduates and scale-ups; routes for founders were plans in Korea and Japan.
- Finance came as vouchers, calls and tax proposals; the United Kingdom will cut its SME R&D relief but extend relief to data and cloud costs and keep larger seed relief.
- Culture received speeches, festivals, rankings and proclamations rather than programmes, and governments reported mostly outputs and claimed effects, with few evaluations.

Digital entrepreneurship depends on data, computing power, rules on data, payments and platforms, and markets reached through platforms that others own. Digital technologies, it has been argued, make entrepreneurial processes and outcomes less bounded (Nambisan, 2017), and the governance of digital infrastructure is itself a domain of policy (Sussan & Acs, 2017). This report applies the entrepreneurial ecosystem approach (Isenberg, 2011; Stam, 2015) to the digital entrepreneurship policy of Korea, Japan, China, the United Kingdom, the United States, France and Germany in 2022. It asks three questions:

1. What did each government set out in each domain of digital entrepreneurship in 2022?
2. How did the policies arise and how were they justified, how far were they valid in theory, and what had they achieved in practice by the end of the year?
3. What were their limitations, and what follows for policy?

The report draws on policy studies and entrepreneurship research where it bears on a policy — on problem representation (Bacchi, 2009; Rochefort & Cobb, 1993), digital ecosystems (Nambisan, Wright, & Feldman, 2019; Sussan & Acs, 2017), platforms (Cutolo & Kenney, 2021), data regulation (Jia, Jin, & Wagman, 2021), entry regulation (Klapper, Laeven, & Rajan, 2006), grants and tax relief (Agrawal, Rosell, & Simcoe, 2020; Howell, 2017), procurement (Edler & Georghiou, 2007) and

immigrant talent (Hunt, 2011). Sections 2 to 5 review the policies by domain and country, analyse them, set out their limits and draw implications.

The report rests on the plans, laws, budgets, reports and announcements of these governments in 2022, including decisions of financial regulators and competition authorities (Table 1). Quotations from sources that are not in English are IFEED's translations.

***Table 1: Government plans, laws, reports and announcements reviewed***

| Subject | Issuing bodies | Date and type | Period covered | Description |
|---|---|---|---|---|
| Korea | | | | |
| Cloud and network separation | FSC | 14 April 2022 Plan | April–December 2022 | Easier cloud use; staged relaxation from 2023. |
| Hyperscale AI pilot | MSIT | 16 June 2022 Press release | May–August 2022 | Model opened to researchers; SMEs later. |
| Digital Asset Task Force | FSC | 17 August 2022 Press release | August–December 2022 | Digital Asset Basic Act study. |
| Digital talent plan | MOE et al. | 22 August 2022 Joint release | August–December 2022 | Training targets by level; school informatics. |
| MSS budget proposal | MSS | 31 August 2022 Budget proposal | August–December 2022 | Proposal; programme with global firms. |
| New York Initiative | MSIT | 22 September 2022 Press release | September 2022 | President's digital vision speech. |
| Digital Strategy of Korea | MSIT | 28 September 2022 Strategy | September–December 2022 | Basic law, visas, exports, culture tasks. |
| Industrial sandbox | MOTIE | 20 December 2022 Plan, press release | January 2019–December 2022 | Approvals, results and plan. |
| Fintech support plan | FSC | 26 December 2022 Plan | December 2022 | Fund, K-Fintech 30, exports. |
| Data Industry Act decree | MSIT | 27 December 2022 Press release | April–December 2022 | Quality certification; fines. |
| MSS work plan; manufacturing act | MSS | 27 December 2022 Press releases | December 2022 | Digital founders, COMEUP; act passed. |
| AI Voucher call | MSIT | 28 December 2022 Press release | December 2022 | Vouchers for buyers of AI. |
| Public data plan | MOIS | 29 December 2022 Press release | December 2022 | Open-by-default data; basic plan. |
| Japan | | | | |
| Payment Services Act amendment | FSA | 4 March 2022 Bill | March–December 2022 | Stablecoin registration; Act No. 61. |
| AI strategy | Innovation Strategy Council | 22 April 2022 Strategy | April–December 2022 | Soft law; AI education target. |
| Innovation strategy | Cabinet | 3 June 2022 Cabinet decision | June–December 2022 | Digital start-ups; Fugaku. |
| Digital society plan | Digital Agency | 7 June 2022 Plan | June–December 2022 | Business services, SME security, procurement. |

| Subject | Issuing bodies | Date and type | Period covered | Description |
|---|---|---|---|---|
| New Capitalism plan | Cabinet | 7 June 2022 Cabinet decision | June–December 2022 | Platforms, crypto screening, AI projects. |
| FSA policy | FSA | August 2022 Policy statement | July 2021–December 2022 | Stablecoins, crypto, fintech desk. |
| Start-up five-year plan | New Capitalism Council | 28 November 2022 Plan | November–December 2022 | Visa reform, Web3.0, procurement. |
| Supplementary budget (draft) | Cabinet Office | November 2022 Budget outline | November–December 2022 | Digital Garden City grant. |
| Digital principles roadmap | Digital Agency | 21 December 2022 Council material | June–December 2022 | Analogue rules; bundle bill planned. |
| Mobile ecosystem issues | Cabinet Secretariat | 22 December 2022 Working group material | April–December 2022 | App stores; ex-ante rules studied. |
| Digital Garden City strategy; tax outline | Cabinet | 23 December 2022 Cabinet decisions | December 2022 | Talent target; crypto and DX tax. |
| Draft payment orders | FSA | 26 December 2022 Press release | December 2022 | Orders under Act No. 61; drafts. |
| China | | | | |
| Computing hub | NDRC et al. | 7 February 2022 Reply | February–December 2022 | Data-centre cluster; targets. |
| Government Work Report | State Council | 5 March 2022 Work report | March–December 2022 | Digital economy; platform fees. |
| Digital government | State Council | 6 June 2022 Opinions | June–December 2022 | Services to firms; digital literacy. |
| Outbound data transfers | CAC | 7 July 2022 Rules | September–December 2022 | Assessment above volume thresholds. |
| AI scenario opinions | MOST et al. | 29 July 2022 Opinions | July–December 2022 | AI start-ups; compute vouchers. |
| SME digital pilot | MIIT; MOF | 15 August 2022 Notice | August–December 2022 | Central rewards to service platforms. |
| SME Digital Service Festival | MIIT | 25 August 2022 Notice | August–September 2022 | Campaign for SMEs' digital adoption. |
| SME digital guide | MIIT | 3 November 2022 Notice, guide | November–December 2022 | Steps, loans, training. |
| Cross-border e-commerce zones | State Council | 14 November 2022 Reply | November–December 2022 | New pilot zones for online trade. |
| Deep synthesis rules | CAC et al. | 25 November 2022 Rules | November–December 2022 | Identity, labels, filing; from 2023. |
| Data 20 Articles | CPC Central Committee; State Council | 2 December 2022 Opinions | December 2022 | Data rights, trading, finance. |
| Industry data security | MIIT | 8 December 2022 Measures | December 2022 | Data grading; filing; from 2023. |

| Subject | Issuing bodies | Date and type | Period covered | Description |
|---|---|---|---|---|
| United Kingdom | | | | |
| Cryptoasset plan | HM Treasury | 4 April 2022 Press release, response | April 2022 | Stablecoins; sandbox; promotion. |
| Queen's Speech | Prime Minister's Office | 10 May 2022 Briefing notes | May–December 2022 | Digital markets and data bills. |
| Procurement Bill | Cabinet Office | 12 May 2022 Press release | May–December 2022 | Small suppliers; digital platform. |
| Global Talent evaluation | Home Office | 30 May 2022 Report | May 2022 | Visa holders' experience. |
| UK Digital Strategy | DCMS | 13 June 2022 Strategy | June–December 2022 | Visas, adoption scheme, stance. |
| AI policy paper; AI Action Plan | DCMS; BEIS; Office for AI | 18 July 2022 Policy paper, plan | July–December 2022 | Non-statutory approach; AI measures. |
| Autumn Statement | HM Treasury | 17 November 2022 Fiscal statement | November 2022 | R&D relief; digital markets bill. |
| Cyber resilience rules | DCMS | 30 November 2022 Response | January–November 2022 | Managed services; small-firm exemption. |
| Product Security Act | Parliament | 6 December 2022 Act | December 2022 | Security rules for connectable products. |
| Edinburgh Reforms | HM Treasury | 9 December 2022 Press release | December 2022 | Crypto exemption; sandbox. |
| Help to Grow: Digital | BEIS | 15 December 2022 Press release | December 2022 | Closure of the software vouchers. |
| United States | | | | |
| STEM degree list | ICE | 21 January 2022 Notice | January–December 2022 | Cloud and data fields for practical training. |
| Cybersecurity pilot | SBA | 24 January 2022 Press release | January–March 2022 | Grants to states for small firms. |
| Executive Order 14067 | President | 9 March 2022 Executive order | March–December 2022 | Digital-asset objectives; reports. |
| CIRCIA | Congress | 15 March 2022 Act | March–December 2022 | Incident reports; rule to follow. |
| Small Business Week | President | 29 April 2022 Proclamation | May 2022 | Broadband for small firms. |
| NAIRR consultation | NSF; OSTP | 25 May 2022 Notice | May–June 2022 | Shared computing for researchers. |
| CHIPS and Science Act | Congress | 9 August 2022 Act | August–December 2022 | NIST AI work; NSF directorate; fellowships. |
| AI risk framework (draft) | NIST | 18 August 2022 Draft framework | August–September 2022 | Voluntary; small organisations. |
| Cyber pilot awards | SBA | 23 August 2022 Press release | August 2022 | Awards to state delegates. |

| Subject | Issuing bodies | Date and type | Period covered | Description |
|---|---|---|---|---|
| Secure software (M-22-18) | OMB | 14 September 2022 Memorandum | September–December 2022 | Attestation for software sold to agencies. |
| Executive Order 14083 | President | 15 September 2022 Executive order | September–December 2022 | Investment review: AI, data. |
| Payments and crypto reports | Treasury | 16 September 2022 Reports | March–September 2022 | Payments framework; existing authorities. |
| Chip export controls | BIS | 13 October 2022 Interim final rule | October–December 2022 | Advanced chips; China. |
| Entrepreneurship Month | President | 31 October 2022 Proclamation | November 2022 | Broadband and online business. |
| AI Bill of Rights Blueprint | OSTP | October 2022 Framework | October 2022 | Five non-binding principles. |
| Cyber Training Act | Congress | 27 December 2022 Act | December 2022 | Cyber counselling certification. |
| France | | | | |
| French Tech Next40/120 | Ministry of the Economy | 1 February 2022 Press kit | February 2022 | Ranking of leading start-ups. |
| Platform cybersecurity audit | Parliament | 3 March 2022 Act | March–December 2022 | Audit of large platforms from 2023. |
| Digital minister's remit | President | 29 July 2022 Decree | July–December 2022 | Remit includes French digital firms. |
| Electronic invoicing act | Parliament | 16 August 2022 Act | August–December 2022 | Electronic invoices between firms. |
| France Num barometer | DGE | 20 September 2022 Press release | September 2022 | Small firms' views; digital cheque. |
| Electronic invoicing decree | Prime Minister | 7 October 2022 Decree | October–December 2022 | Partner platforms; public contracts. |
| 5G strategy | MEFIDS | 12 October 2022 Press release | March–October 2022 | Frequency fees; test platforms. |
| Trusted cloud partnership | MEFIDS | 26 October 2022 Press release | October 2022 | Sovereign cloud offer; call to buyers. |
| Cybersecurity projects | MEFIDS | 27 October 2022 Press release | February 2021–October 2022 | EUR 39 million for 17 projects. |
| DeepNum20 | French Tech Mission | 27 October 2022 Press kit | July–October 2022 | Deep-tech digital start-ups supported. |
| Council on digital | MEFIDS | 22 November 2022 Press release | November 2022 | Consultation on citizens' digital life. |
| SecNumCloud scheme | MEFIDS | 22 December 2022 Press release | December 2022 | EUR 3.5 million scheme. |
| DGE strategy | DGE | 2022 Strategy | 2022 | Digital transition aims; France Num. |
| Germany | | | | |
| Google decision | Federal Cartel Office | 5 January 2022 Press release | January 2022 | Section 19a determination. |

| Subject | Issuing bodies | Date and type | Period covered | Description |
|---|---|---|---|---|
| Annual Economic Report | Federal Government | 26 January 2022 Report | January–December 2022 | Data access; stipends for women. |
| Meta decision | Federal Cartel Office | 4 May 2022 Press release | May 2022 | Section 19a determination. |
| Apple proceeding | Federal Cartel Office | 14 June 2022 Press release | June 2022 | App tracking rules examined. |
| Amazon decision | Federal Cartel Office | 6 July 2022 Press release | July 2022 | Section 19a determination. |
| Start-up Strategy summary | Press Office | 27 July 2022 Article | July 2022 | Summit; contact points; portal. |
| Start-up Strategy | Federal Government | 28 July 2022 Strategy | July–December 2022 | Laboratories, open data, formation. |
| Draft budget for 2023 | Federal Government | 5 August 2022 Budget bill | August–December 2022 | Digital agenda; AI service centres. |
| Digital Strategy | Federal Government | 5 September 2022 Strategy | August–December 2022 | Priorities; AI; financing act. |
| Online Access Act status | Federal Government | 4 November 2022 Answer | April–November 2022 | Online services; digital check. |
| Skilled immigration key points | Federal Government | 30 November 2022 Key points | June–November 2022 | Blue Card; IT specialists. |
| SME innovation report | Federal Government | 6 December 2022 Report | January 2021–December 2022 | Digital Now, go-digital, competition. |

Source: The plans, laws, reports and announcements listed. Periods covered and descriptions: IFEED.

Note: Where a source has several parts, citations name the part (for example, plan or press release) before the page. Abbreviations: AI, artificial intelligence; ANSSI, National Cybersecurity Agency of France; BEIS, Department for Business, Energy and Industrial Strategy; BIS, Bureau of Industry and Security; BMBF, Federal Ministry of Education and Research; BMWK, Federal Ministry for Economic Affairs and Climate Action; CAC, Cyberspace Administration of China; CFIUS, Committee on Foreign Investment in the United States; CHIPS, Creating Helpful Incentives to Produce Semiconductors; CIRCIA, Cyber Incident Reporting for Critical Infrastructure Act; CISA, Cybersecurity and Infrastructure Security Agency; CP, Command Paper; CPC, Communist Party of China; DCMS, Department for Digital, Culture, Media and Sport; DGE, Directorate-General for Enterprises; DX, digital transformation; EU, European Union; FSA, Financial Services Agency; FSC, Financial Services Commission; HM, Her or His Majesty's; ICE, U.S. Immigration and Customs Enforcement; ICT, information and communications technology; ID, identification; IPO, initial public offering; IT, information technology; MEFIDS, Ministry of the Economy, Finance and Industrial and Digital Sovereignty; MIIT, Ministry of Industry and Information Technology; MOE, Ministry of Education; MOF, Ministry of Finance; MOIS, Ministry of the Interior and Safety; MOST, Ministry of Science and Technology; MOTIE, Ministry of Trade, Industry and Energy; MSIT, Ministry of Science and ICT; MSS, Ministry of SMEs and Startups; NAIRR, National Artificial Intelligence Research Resource; NDRC, National Development and Reform Commission; NIST, National Institute of Standards and Technology; NSF, National Science Foundation; OMB, Office of Management and Budget; OSTP, Office of Science and Technology Policy; R&D, research and development; SBA, Small Business Administration; SME, small and medium-sized enterprise; STEM, science, technology, engineering and mathematics; UK, United Kingdom; US and U.S., United States; VAT, value added tax.

# 2 REVIEW OF DIGITAL ENTREPRENEURSHIP POLICIES

This section reviews the key policies of 2022 in six domains adapted from Isenberg's domains of the entrepreneurial ecosystem (Isenberg, 2011): regulation and law; finance; culture; institutional support; human capital; and markets, including initial public offerings and exit. In digital entrepreneurship they cover rules on data, crypto-assets, AI and platforms; money for adoption, security and cloud; digital services; routes for talent; and public demand and platform power.

## 2.1 REGULATION AND LAW

Regulation turned on data and payments: binding data rules in China, a stablecoin law in Japan, non-binding frameworks in the United States and the United Kingdom, invoicing and audit laws in France, and plans in Korea and Germany.

### 2.1.1 KOREA: THE DATA INDUSTRY ACT DECREE AND THE INDUSTRIAL SANDBOX

On 27 December the Cabinet approved an amendment of the enforcement decree of the Data Industry Act, in force since April, which sets requirements for data quality-certification bodies and fines of KRW 5 million to KRW 10 million for false certification marks; it will take effect on 3 January 2023 (MSIT, 27 December 2022, paras 1–2 and item 4). MOTIE's industrial convergence sandbox had approved 327 projects since January 2019, 129 of them in 2022; 173 approved firms had started business, with KRW 163.1 billion of sales and 866 jobs by MOTIE's count, and the 33 regulations behind 51 tested projects had been revised. Its plan of 20 December would add a KRW 50 billion fund, and MOTIE would consider a special act suspending old rules while they are revised (MOTIE, 20 December 2022, summary and parts 1–2). The SME Smart Manufacturing Innovation Promotion Act passed the National Assembly on 8 December and the Cabinet on 27 December, and MSS said that it will be promulgated on 3 January 2023 and take effect six months later (MSS, 27 December 2022, Act release, remarks and reference 1). The Digital Strategy of 28 September planned to enact an AI Basic Act, among five foundational laws of the digital economy, and a Digital Society Basic Act (MSIT, 28 September 2022, strategy, pp. 68, 91), and the FSC planned in April to ease the cloud and network-separation rules for financial firms from 2023 (FSC, 14 April 2022, summary and future plan).

This was a mixed policy. The sandbox works: tests under exemption changed the rules for 51 projects, so that any firm may now offer them, and the governance of rules and infrastructure, it has been argued, shapes where digital ventures can enter (Sussan & Acs, 2017). But the Smart Manufacturing Act serves manufacturers' digital adoption rather than digital ventures, and the basic acts, the special act and the financial cloud rules were plans.

### 2.1.2 JAPAN: THE STABLECOIN LAW AND THE DIGITAL PRINCIPLES ROADMAP

The amendment of the Payment Services Act, passed on 3 June and promulgated on 10 June as Act No. 61, makes the handling of electronic payment instruments, the stablecoins of the Act, a business registered with the Prime Minister, and will take effect on a date set by Cabinet Order within one year of promulgation; FSA published draft implementing orders on 26 December (FSA, 4

March 2022, pp. 24, 223; FSA, 26 December 2022, p. 1). Japan had no AI statute: AI Strategy 2022 of 22 April rests on the social principles of human-centric AI of 2019 and seeks standards that do not unduly constrain economic activity (Integrated Innovation Strategy Promotion Council, 22 April 2022, pp. 2, 40). On 21 December the Digital Agency's council fixed a roadmap for reviewing about 10,000 provisions of analogue regulation and planned a bundle bill for the ordinary Diet session of 2023 (Digital Agency, 21 December 2022, pp. 11, 17), and FSA undertook to speed the screening of new crypto-assets and to clarify when blockchain tokens are crypto-assets (FSA, August 2022, p. 32).

This was a mixed policy. The stablecoin law gives issuers and intermediaries a licensed route, and the roadmap targets rules requiring paper, presence or inspection by eye, whose removal the Digital Agency expects to help start-ups rise (Digital Agency, 21 December 2022, p. 2). Principles without duties spared AI start-ups new compliance costs. But the law's orders were drafts, and the bundle bill and the crypto screening were plans.

### 2.1.3 CHINA: OUTBOUND DATA MEASURES AND THE DEEP SYNTHESIS PROVISIONS

The CAC's measures for the security assessment of outbound data transfers, in force from 1 September, require an assessment for any export of important data, for operators of critical information infrastructure and processors of the personal information of more than 1 million people, and for processors that have exported the personal information of 100,000 people, or the sensitive personal information of 10,000, since 1 January of the previous year (CAC, 7 July 2022, Arts. 4, 20). The deep synthesis provisions of 25 November, issued by the CAC with MIIT and the Ministry of Public Security, will take effect on 10 January 2023 and will require providers to verify users' identity, assess tools that generate faces or voices, label synthetic content and file their algorithms (CAC, MIIT, & Ministry of Public Security, 25 November 2022, Arts. 9, 15–19, 25), and MIIT's data security measures for industry will apply from 1 January 2023, with important and core data to be filed and assessed every year (MIIT, 8 December 2022, Arts. 12, 31, 42). The Data 20 Articles of 2 December, issued by the CPC Central Committee and the State Council, set the direction for separate rights to hold, use and trade data and asked platform enterprises to authorise data fairly in both directions with small and micro firms (CPC Central Committee & State Council, 2 December 2022, Part II, arts. (3) and (5)).

This was a mixed policy. The volume thresholds spare small exporters an assessment, a fixed cost of the kind that, under uniform data rules, has weighed most on young, data-related ventures (Jia et al., 2021), and fair data authorisation would open platform data to small firms. But the duties of the deep synthesis rules will apply whatever a provider's size, and the Data 20 Articles set a direction whose rules were still to come.

### 2.1.4 UNITED KINGDOM: THE AI POLICY PAPER AND THE PRODUCT SECURITY ACT

The policy paper of 18 July (CP 728) proposed to regulate AI by its use and context through existing regulators, with principles at first on a non-statutory footing; the government saw no need for legislation "at this stage", noted that unclear rules hurt smaller businesses, and planned a white paper for late 2022 (DCMS, BEIS, & Office for AI, 18 July 2022, policy paper, pp. 2, 6, 17–18). The

Queen's Speech of 10 May announced a Data Reform Bill to reduce burdens on businesses and widen Smart Data Schemes (Prime Minister's Office, 10 May 2022, p. 57). The Product Security and Telecommunications Infrastructure Act 2022, which received Royal Assent on 6 December, lets the Secretary of State set security requirements for connectable consumer products by regulations; most of it commences by regulations (Parliament of the United Kingdom, 6 December 2022, ss. 1(1), 38(1), 79(2)). On 30 November DCMS decided to bring managed service providers under the network and information systems regulations largely keeping the small and micro business exemption, subject to finding a legislative vehicle (DCMS, 30 November 2022, sections 2.3 and 4.1).

This was a mixed policy. The non-statutory approach and the small-firm exemption spare small firms new compliance costs, and the Act gives makers of connected products one set of security rules. But the AI approach and the data reform were proposals at the end of the year, the cyber rules lacked a vehicle, and the Act's duties await regulations.

### 2.1.5 UNITED STATES: EXECUTIVE ORDER 14067 AND CIRCIA

Executive Order 14067 of 9 March set the objectives of policy on digital assets, and the principle "same business, same risks, same rules" for digital-asset firms (President of the United States, 9 March 2022, secs 2(a)–(b)); Treasury's reports of 16 September asked regulators to use existing authorities while Congress considered legislation, and proposed a federal framework for regulating non-bank payment firms (Department of the Treasury, 16 September 2022, crypto-asset report, p. 54, and payments report, p. 51). OSTP's Blueprint for an AI Bill of Rights of October set out five principles and stated that it "is non-binding" (OSTP, October 2022, pp. 2–3), and NIST's second draft of its voluntary AI Risk Management Framework, of 18 August, noted that small and medium-sized organisations may face challenges different from those of large ones (NIST, 18 August 2022, pp. 1, 15). The Cyber Incident Reporting for Critical Infrastructure Act, enacted on 15 March as Division Y of Public Law 117-103, requires covered entities to report covered incidents to CISA within 72 hours and ransom payments within 24 hours once a final rule sets the dates; rulemaking was to follow (United States Congress, 15 March 2022, Division Y, sec. 103).

This was a mixed policy. The order and the AI frameworks imposed no new duties on young firms, and NIST named the needs of small organisations. But the order directed reports rather than rules, the AI frameworks bound no one, the payments framework was a recommendation, and CIRCIA's duties await a rule.

### 2.1.6 FRANCE: ELECTRONIC INVOICING AND THE PLATFORM SECURITY AUDIT

Article 26 of the amending finance law of 16 August (Law No. 2022-1157) obliges VAT-liable businesses established in France to issue, transmit and receive invoices electronically through the public invoicing portal or another dematerialisation platform, and Decree No. 2022-1299 of 7 October sets the rules for partner platforms; the obligation will be phased in from a later date, with SMEs and micro-enterprises last (Parliament of France, 16 August 2022, Art. 26; Prime Minister of France, 7 October 2022, Arts. 1, 3). Law No. 2022-309 of 3 March will require online platforms and interpersonal communication services above thresholds set by decree to have their cybersecurity audited by providers qualified by ANSSI and to show the result to consumers, from 1 October 2023

(Parliament of France, 3 March 2022, Arts. 1–2). The DGE aimed to carry the EU's rules on digital markets and services further in French law (DGE, 2022, p. 10).

This was a mixed policy. Electronic invoicing through partner platforms creates a regulated market for invoicing software and will cut the cost of compliance for small firms once in use, and the audit duty falls only on platforms above thresholds. But the invoicing and audit duties did not yet apply, and the national follow-up of the EU's rules on digital markets was an aim.

### 2.1.7 GERMANY: THE REAL-WORLD LABORATORIES ACT AND THE OPEN DATA RIGHT

The Start-up Strategy, adopted by the Cabinet on 27 July, announced a Real-World Laboratories Act with experimentation clauses, a mandatory check for them and a one-stop shop for start-ups, AI real-world laboratories giving start-ups preferential access, and a legal right to the Federation's open data, free and machine-readable (Federal Government of Germany, 28 July 2022, pp. 15–16). On 4 November the government said that a digital check for federal laws was being designed and that the need for a national data law would depend on the negotiations on the EU's data act (Federal Government of Germany, 4 November 2022, pp. 3, 5).

This was a mixed policy. Real-world laboratories with preferential access for start-ups and a right to open data address the costs of testing products and finding data, and open public data are an input that, it has been argued, digital ventures recombine (Nambisan et al., 2019). But each was announced, not tabled, at the end of the year, and the data law waited on the EU's negotiations.

## 2.2 FINANCE: GRANTS, FUNDING AND FINANCIAL INFRASTRUCTURE

Finance took the form of vouchers and subsidies for digital adoption, tax rules for crypto-assets and research, calls for cybersecurity and cloud, and budget lines in drafts and proposals.

### 2.2.1 KOREA: THE AI VOUCHER PROGRAMME AND THE FINTECH INNOVATION FUND

On 28 December MSIT opened the call for its AI Voucher programme for 2023, under which SMEs, mid-sized firms and medical institutions buy AI solutions from a pool of domestic AI suppliers; it plans to select about 250 projects with KRW 70 billion, vouchers are worth up to KRW 300 million and SMEs bear 20 % of the cost; 1,573 projects applied in 2022 (MSIT, 28 December 2022, paras 1–3, box and attachment 1). On 26 December the FSC announced that the Fintech Innovation Fund, KRW 500 billion over four years, would receive a further KRW 500 billion over the following four years (FSC, 26 December 2022, part 1). MSS's work plan of 27 December promised up to KRW 100 million of commercialisation funding for prospective and early founders in AI, data and other digital technologies (MSS, 27 December 2022, work plan release, part 1(2)), and the FSC's task force, launched on 17 August, began to study a Digital Asset Basic Act (FSC, 17 August 2022, summary box).

This was a mixed policy. The vouchers create paying customers for domestic AI suppliers, which MSIT expects to be small and pioneering firms, and the founder funding aims at AI and data start-ups directly; early-stage grants have been found to raise later venture funding, most for financially

constrained firms (Howell, 2017). But no project had yet been selected under the call, the founder funding and the fintech top-up were announcements, and the digital-asset law was under study.

### 2.2.2 JAPAN: THE CRYPTO-ASSET TAX RELIEF AND THE DX INVESTMENT MEASURE

The tax reform outline for fiscal year 2023, a Cabinet decision of 23 December, would exclude from year-end mark-to-market taxation crypto-assets that a corporation has issued itself and held continuously under transfer restrictions, as the Start-up Development Five-year Plan had promised for Web3.0 firms (Cabinet of Japan, 23 December 2022, tax outline, p. 54; New Capitalism Realisation Council, 28 November 2022, p. 17). The outline listed the DX investment tax measure among measures "abolished or reduced": it would be extended by two years but approved only for plans expecting sales growth of at least 10 % and a set share of overseas sales (Cabinet of Japan, 23 December 2022, tax outline, p. 47). The crypto-asset relief and the DX measure will need legislation. The Cabinet Office's draft second supplementary budget of November included JPY 80.0 billion for the Digital Garden City Nation grant (Cabinet Office of Japan, November 2022, p. 3), whose selection gives a bonus to start-up projects (Cabinet of Japan, 23 December 2022, strategy, p. 9).

This was a mixed policy. Relief for self-issued crypto-assets removes a tax on unrealised gains on tokens that Web3.0 firms hold to run their business, and the grant rewards start-up projects. But the reliefs were outline proposals, the DX measure was narrowed, and the grant was a draft whose money goes to local governments rather than to firms.

### 2.2.3 CHINA: THE FISCAL PILOT FOR SME DIGITAL TRANSFORMATION

Under the notice of 15 August of the General Offices of MIIT and MOF, the central budget rewards digital service platforms that complete the transformation of pilot SMEs, mainly in manufacturing, with up to CNY 6 million per platform, calculated at no more than 30 % of each firm's actual cost and CNY 300,000 per firm; about 100 platforms were to be supported in 2022, and the pilot is planned in three batches (General Offices of MIIT & MOF, 15 August 2022, Parts I and III). The guide of MIIT's General Office of 3 November asked banks for special loans for SMEs' digital transformation (General Office of MIIT, 3 November 2022, pp. 6–7), and the Data 20 Articles asked for venture capital to be guided into data-factor firms and for credit based on firms' operating data (CPC Central Committee & State Council, 2 December 2022, Part VI, art. (18)).

This was a mixed policy. The pilot puts central money into SMEs' digital transformation, makes firms and platforms bear part of the cost, and pays the digital service platforms that do the work. But it serves established manufacturers rather than new ventures, and the venture capital and credit for data firms were guidance without money.

### 2.2.4 UNITED KINGDOM: R&D RELIEF FOR DATA AND CLOUD COSTS

The Autumn Statement of 17 November kept the announced increases to seed-investment relief and company share options for innovative firms (HM Treasury, 17 November 2022, para. 3.31), confirmed that R&D reliefs would cover data and cloud costs, to be legislated in the Spring Finance Bill 2023, and that from 1 April 2023 the SME additional deduction will fall from 130 % to 86 % and

the SME credit rate from 14.5 % to 10 %, while the R&D expenditure credit rises from 13 % to 20 % (HM Treasury, 17 November 2022, para. 5.52). On 4 April HM Treasury said that it would legislate to bring payment stablecoins into the payments perimeter and consult on wider cryptoasset activity (HM Treasury, 4 April 2022, press release), and on 9 December it said that the Investment Management Exemption would be extended to cryptoassets (HM Treasury, 9 December 2022, section on innovation and technology). The AI Action Plan of 18 July announced GBP 35 million for Innovate UK programmes on AI adoption (DCMS et al., 18 July 2022, action plan, Pillar Two).

This was a mixed policy. Counting data and cloud costs as R&D brings AI firms' main costs into the relief, larger seed relief and share options help start-ups raise money and hire, and the crypto plans offer a route into regulation. But the SME relief will be cut, although refundable credits have been found to raise the R&D of small firms, most for firms with no tax to pay (Agrawal et al., 2020), and the crypto measures were plans.

### 2.2.5 UNITED STATES: THE CYBERSECURITY FOR SMALL BUSINESS PILOT

In January SBA announced USD 3 million for state governments under its Cybersecurity for Small Business Pilot Program, with grants of up to USD 1 million, to help emerging small businesses build their cybersecurity, and on 23 August it awarded nearly USD 3 million to bodies in Arkansas, Maryland and South Dakota (SBA, 24 January 2022, paras 1, 4; SBA, 23 August 2022, para. 1 and list of awards). Treasury's report of 16 September proposed a federal framework for payments regulation that would cover non-bank payment firms, whose participation, it noted, may raise competition, inclusion and innovation (Department of the Treasury, 16 September 2022, payments report, p. 51).

This was a mixed policy. The pilot pays for training, counselling and remediation for nascent and start-up firms. But it was small, its money went to bodies chosen by the states rather than to firms, and the payments framework for fintech entrants was a recommendation.

### 2.2.6 FRANCE: THE CYBERSECURITY AND CLOUD CALLS OF THE INVESTMENT PLAN

On 27 October the minister for the digital transition announced EUR 39 million of public funding for 17 projects selected through five calls of the national cybersecurity strategy, which allocates more than EUR 1 billion, EUR 720 million of it public (MEFIDS, 27 October 2022, pp. 1–2). On 22 December the government opened, through Bpifrance, a EUR 3.5 million scheme financed by the investment plan's cloud strand to pay for audits, advice and aid towards the SecNumCloud security qualification for start-ups and SMEs; the strand, endowed with EUR 667 million, supports emerging players, SMEs and start-ups (MEFIDS, 22 December 2022, pp. 1–2).

This was a mixed policy. Competitive calls are the kind of support whose effect has been found largest for financially constrained firms (Howell, 2017), and the SecNumCloud scheme pays start-ups and SMEs for a qualification that the government itself called demanding and costly (MEFIDS, 22 December 2022, p. 1). But no money from the scheme had reached firms by the end of the year, and the 17 projects spanned training and shared data as well as firms.

### 2.2.7 GERMANY: DIGITAL BUDGET LINES AND THE FUTURE FINANCING ACT

The government's draft budget for 2023 of 5 August put the BMWK's digital agenda funding area at about EUR 1.1 billion, including about EUR 98 million for Digital Now, the investment grant for SMEs' digitalisation (Federal Government of Germany, 5 August 2022, p. 1069), and raised the BMBF's line for IT infrastructure for AI service centres, which are also to support SMEs, from EUR 18.0 million in 2022 to EUR 24.0 million (Federal Government of Germany, 5 August 2022, p. 3044). The Start-up Strategy and the Digital Strategy named a planned Future Financing Act as the main step to ease initial public offerings and to mobilise private capital for growth finance (Federal Government of Germany, 28 July 2022, p. 4; Federal Government of Germany, 5 September 2022, p. 25).

This was a mixed policy. The draft raised the money for AI service centres open to SMEs and kept a grant for SMEs' digital investment. But the lines were proposals in a draft budget, and the act meant to open capital markets to growth firms was a plan.

## 2.3 CULTURE

Culture was the thinnest domain: speeches, conferences, festivals, recognition schemes and proclamations, and few programmes that change attitudes to founding digital firms.

### 2.3.1 KOREA: THE NEW YORK INITIATIVE, COMEUP AND K-FINTECH 30

On 21 September the President gave a keynote at a digital vision forum of New York University on "solidarity for digital free citizens", setting out a vision of a digital order based on freedom, human rights and solidarity (MSIT, 22 September 2022, paras 4–5). MSS planned to grow COMEUP, its start-up festival, into one of the world's five largest start-up events (MSS, 27 December 2022, work plan release, part 1(1)), and the FSC planned to select 30 leading fintech firms as "K-Fintech 30" (FSC, 26 December 2022, part 1).

This was a mixed policy. A festival that brings foreign start-ups to Korea and the selection of leading fintech firms build the networks and role models through which, it has been argued, an entrepreneurial culture is reproduced (Spigel, 2017). But COMEUP's growth and K-Fintech 30 were plans, and the President's initiative addressed a digital world order rather than founders.

### 2.3.2 JAPAN: FSA'S FINTECH CONFERENCE AND THE START-UP DECLARATION

FSA's fintech conference of 29 to 31 March brought together fintech firms, financial institutions, academics and officials from Japan and abroad (FSA, August 2022, p. 176). The Start-up Development Five-year Plan of 28 November declared 2022 the "first year of start-up creation" and aimed at a second founding boom (New Capitalism Realisation Council, 28 November 2022, p. 1), and the Integrated Innovation Strategy of 3 June aimed for deep-tech and digital start-ups to arise one after another (Cabinet of Japan, 3 June 2022, p. 6). The Digital Agency's priority plan of 7 June used the DX certification and a DX brand for listed firms to push digital transformation through market evaluation (Digital Agency, 7 June 2022, p. 91).

This was a mixed policy. A conference that gathers fintech firms with regulators builds the relations through which ecosystems, it has been argued, sustain their culture (Spigel, 2017), and the plan made founding a national aim. But the declaration set no programme of its own, and the DX schemes honour established firms rather than founders.

### 2.3.3 CHINA: THE NATIONAL SME DIGITAL SERVICE FESTIVAL

MIIT's General Office scheduled a national SME Digital Service Festival from 28 August to 28 September, aimed at SMEs that are "unwilling, not knowing how, afraid or unable" to transform digitally, with support in technology, experience, sales channels and capital, including special financing products (General Office of MIIT, 25 August 2022, Parts I and III).

This was a mixed policy. A national campaign reached SMEs reluctant to adopt digital tools and brought digital service providers to them. But it promoted adoption rather than founding, set out no role models or networks for digital founders (Spigel, 2017), and reported no results in the year.

### 2.3.4 UNITED KINGDOM: THE UK DIGITAL STRATEGY AND CRYPTO ENGAGEMENT

The UK Digital Strategy of 13 June aimed to make the United Kingdom the best place in the world to start and grow a technology business and held that the success and wealth created by investors and founders of digital business "is a national success, to be applauded, encouraged and emulated – not criticised" (DCMS, 13 June 2022, foreword). On 4 April HM Treasury announced that the Chancellor had commissioned the Royal Mint to create a non-fungible token that summer, and announced a CryptoSprint of the Financial Conduct Authority and a Cryptoasset Engagement Group (HM Treasury, 4 April 2022, press release, other measures).

This was a mixed policy. A stance that celebrates founders, and engagement with crypto firms, build the role models and relations of an entrepreneurial culture (Spigel, 2017). But the stance carried no programme, and the token and the engagement measures were announcements whose delivery was not reported.

### 2.3.5 UNITED STATES: SMALL BUSINESS WEEK AND ENTREPRENEURSHIP MONTH

By Proclamation 10384 of 29 April, for National Small Business Week, the President said that the broadband provisions of the Bipartisan Infrastructure Law would let every small business use digital technologies and gain new customers, and by Proclamation 10487 of 31 October he declared November National Entrepreneurship Month, noting that broadband was coming to small towns and rural areas so that Americans anywhere can run a business online (President of the United States, 29 April 2022, p. 26667; President of the United States, 31 October 2022, p. 66525). Treasury recommended that consumers, investors and businesses have access to trustworthy information on crypto-assets (Department of the Treasury, 16 September 2022, crypto-asset report, p. 57).

This was a poor policy. The proclamations celebrated entrepreneurs and connectivity but set no programme, and the Treasury advice addressed public understanding of crypto-assets, not founders; none built the networks or role models of an entrepreneurial culture (Spigel, 2017).

### 2.3.6 FRANCE: FRENCH TECH NEXT40/120 AND THE FRANCE NUM BAROMETER

On 1 February the government announced the 2022 cohort of French Tech Next40/120, its ranking of leading start-ups, each of which receives an account manager, visibility in official delegations and access to a community for sharing experience (Ministry of the Economy, Finance and Recovery, 1 February 2022, pp. 1, 6). The France Num barometer of 20 September found that 81 % of the heads of very small firms and SMEs saw a real benefit in digital technology, against 78 % in 2021 (DGE, 20 September 2022, p. 1). On 22 November the minister for the digital transition launched the digital strand of the National Council for Renewal, on digital inclusion, work and a calmer online space (MEFIDS, 22 November 2022, p. 1).

This was a mixed policy. The Next40/120 ranking holds up founders as role models and joins them in a community, through which, it has been argued, an entrepreneurial culture is reproduced (Spigel, 2017). But the barometer is a survey, not a programme, and the council addressed citizens' digital life rather than founders.

### 2.3.7 GERMANY: THE FOUNDING COMPETITION AND THE START-UP SUMMIT

The BMWK's founding competition "Digital Innovations" supports new firms in the digital economy and honoured 21 winning teams in each of its two rounds of 2021(Federal Government of Germany, 6 December 2022, p. 3). On 27 July the government announced that it would organise a first Start-up Summit Germany and set up contact points for start-ups in all federal ministries (Press and Information Office of the Federal Government, 27 July 2022, section on networking), and its Start-up Strategy held that founders deserve social and political appreciation for taking risks (Federal Government of Germany, 28 July 2022, p. 2). The Annual Economic Report of 26 January aimed to raise the share of women founders in the digital sector through stipends (Federal Government of Germany, 26 January 2022, p. 68).

This was a mixed policy. A standing competition that honours digital founders, and a summit with them, build the role models and networks of an entrepreneurial culture (Spigel, 2017). But the competition's reported results were from 2021, and the summit, the contact points and the stipends were announcements.

## 2.4 SUPPORT: INSTITUTIONAL SUPPORT

Support turned on SME digitalisation, public digital services for firms and institutions for data and AI; access to computing for start-ups was rare.

### 2.4.1 KOREA: THE HYPERSCALE AI PILOT AND THE PUBLIC DATA PLAN

In June MSIT began a pilot opening Naver Cloud's hyperscale AI model HyperCLOVA, through an interface, to public and research bodies and universities, and planned to consider extending it to SMEs after a trial to August (MSIT, 16 June 2022, paras 1, 5). The Public Data Strategy Committee, launched on 29 December, discussed a basic plan for public data of KRW 1.1925 trillion over three years and a move to open all public data unless an agency justifies withholding it (MOIS, 29 December 2022, summary box and section on the basic plan). The FSC planned a council of the

bodies that run fintech programmes and expert teams for early-stage firms (FSC, 26 December 2022, parts 1–2), and MSS a virtual "digital venture valley" for start-ups (MSS, 27 December 2022, work plan release, part 1(2)).

This was a mixed policy. Opening a large AI model and public data lowers the cost of the models and data on which AI ventures build, inputs that, it has been argued, a digital entrepreneurial ecosystem must make open (Sussan & Acs, 2017). But the AI pilot served researchers, with SMEs only to be considered, and the data plan, the fintech council and the MSS measures were announcements.

### 2.4.2 JAPAN: G-BIZ ID, THE FINTECH SUPPORT DESK AND SME CYBERSECURITY

The Digital Agency's priority plan of 7 June set improvements, within fiscal year 2022, to G-Biz ID, the common login for business-to-government procedures, and to e-Gov and the J-Grants subsidy system (Digital Agency, 7 June 2022, p. 89), and it promoted an affordable package of security services for SMEs (Digital Agency, 7 June 2022, p. 91). FSA's FinTech Support Desk received 306 consultations in the 2021 administrative year, and its proof-of-concept hub had completed support for three projects and published their results (FSA, August 2022, p. 174). The Digital Garden City Nation strategy of 23 December offered local governments considering Web3.0 a consultation desk (Cabinet of Japan, 23 December 2022, strategy, p. 42), and the supercomputer Fugaku was operated for wide use by academia and industry (Cabinet of Japan, 3 June 2022, p. 103).

This was a mixed policy. One login, online subsidy applications and a fintech desk lower the cost of dealing with the state and with financial rules, and the desk's use was reported. But the login and subsidy-system improvements were plans for the fiscal year, and the Web3.0 desk served local governments rather than firms.

### 2.4.3 CHINA: THE COMPUTING HUB AND THE AI SCENARIO OPINIONS

On 7 February the NDRC and other departments agreed to start building the Beijing–Tianjin–Hebei hub of the national integrated computing network, with a data-centre cluster in Zhangjiakou and targets for rack occupancy of at least 65 % and power usage effectiveness below 1.25 (NDRC, Office of the Central Cyberspace Affairs Commission, MIIT, & National Energy Administration, 7 February 2022, items 1, 3, 5). MOST's opinions of 29 July, with other ministries, named AI start-ups among the firms to be supported in scenario innovation and encouraged localities to cut AI firms' infrastructure costs through sharing, purchased services and innovation vouchers (Ministry of Science and Technology, Ministry of Education, MIIT, Ministry of Transport, Ministry of Agriculture and Rural Affairs, & National Health Commission, 29 July 2022, items 8, 15).

This was a mixed policy. Computing infrastructure and vouchers for computing address the costs of digital transformation, and the AI opinions named start-ups. But the computing hub offered no route for start-ups, and the AI support was an unfunded encouragement to localities.

### 2.4.4 UNITED KINGDOM: THE CLOSURE OF HELP TO GROW: DIGITAL

Help to Grow: Digital offered SMEs up to 50 % of the cost of approved software, up to GBP 5,000, and aimed at up to 100,000 businesses (DCMS, 13 June 2022, section 5.1); on 15 December BEIS announced that it will close to new applications on 2 February 2023 because take-up was lower than expected, with fewer than 1,000 vouchers redeemed (BEIS, 15 December 2022, paras 1–4). The AI Action Plan of 18 July announced a pilot AI Standards Hub led by the Alan Turing Institute and reported a DCMS review of the future of compute under way (DCMS et al., 18 July 2022, action plan, Pillars One and Three), the Digital Strategy planned a Digital Growth Grant competition for the acceleration of start-ups and scale-ups over the summer (DCMS, 13 June 2022, section 4), and HM Treasury said on 4 April that a Financial Market Infrastructure Sandbox would be up and running in 2023 (HM Treasury, 4 April 2022, response, para. 1.8). The Autumn Statement announced that Made Smarter Adoption, which helps smaller manufacturers adopt digital technology in four English regions, would be extended to the East Midlands (HM Treasury, 17 November 2022, para. 3.29).

This was a mixed policy. The standards hub and the sandbox plan address the uncertainty of rules for AI and financial technology, Made Smarter Adoption ran and was to be extended, and the government gave its reasons for closing the voucher scheme. But the main scheme for SMEs' software adoption will close, with Help to Grow: Management remaining and no successor for software vouchers named (BEIS, 15 December 2022, paras 2, 4); the compute review had not reported, and the grant competition and the sandbox were announcements.

### 2.4.5 UNITED STATES: THE NAIRR CONSULTATION AND THE NSF DIRECTORATE

On 25 May NSF and OSTP sought comments, until 30 June, on the interim findings of the NAIRR Task Force, which Congress had directed to plan a shared infrastructure giving AI researchers and students computing, data, tools and support (NSF & OSTP, 25 May 2022, p. 31914). The CHIPS and Science Act of 9 August established NSF's Directorate for Technology, Innovation, and Partnerships and provided for Regional Innovation Engines and a Regional Technology and Innovation Hub Program (United States Congress, 9 August 2022, secs 10381, 10388, 10621), and NIST asked how smaller organisations could use its AI framework (NIST, 18 August 2022, p. 3).

This was a mixed policy. The Act gives research translation and regional technology clusters a statutory base, and NIST sought the views of small organisations. But the research resource was a consultation aimed at researchers and students rather than firms, and no programme gave start-ups computing, data or advice of their own.

### 2.4.6 FRANCE: THE DEEPNUM20 PROGRAMME, FRANCE NUM AND THE 5G PLATFORMS

On 27 October the French Tech Mission announced the first cohort of its DeepNum20 programme for deep-tech digital start-ups, chosen by a jury after ministerial pre-selection, including six start-ups in AI and three each in cybersecurity, space and quantum technology; for a year they receive a start-up manager and access to State services through French Tech correspondents in more than 60 administrations, including the French Tech Visa (French Tech Mission, 27 October 2022, pp. 6–9). The DGE reported that France Num had delivered nearly 150,000 support actions for very small

firms and SMEs since its creation (DGE, 2022, p. 9). Under the 5G acceleration strategy, 21 platforms for testing uses of 5G had been supported (MEFIDS, 12 October 2022, p. 2).

This was a good policy. DeepNum20 reached deep-tech digital start-ups directly, selected by competition, and France Num and the 5G platforms brought digital tools to small and industrial firms. Its limits are that DeepNum20 offers guidance and access to services rather than money, and that its selection by technology is the kind of targeting that, it has been argued, the variety of high-growth firms makes impractical (Mason & Brown, 2013).

### 2.4.7 GERMANY: DIGITAL NOW, GO-DIGITAL AND THE ONLINE ACCESS ACT

Digital Now, the BMWK's grant for SMEs' digital investment, had just under EUR 475 million and had approved more than 2,000 applications by the end of 2021; go-digital, without which nearly 94 % of SMEs surveyed in its 2021 evaluation said they would not have digitalised in the same way, was modernised from January 2022; Mittelstand-Digital informed SMEs about AI through more than 50 AI trainers, and the Digital Hub Initiative linked start-ups with the Mittelstand at 12 locations (Federal Government of Germany, 6 December 2022, pp. 4–5). The Online Access Act obliged the Federation and the Länder to offer their administrative services online by the end of 2022; about 7,600 services were to be digitised, and the government reported on 4 November that 106 bundled services had been implemented in the active programme (Federal Government of Germany, 4 November 2022, p. 2). The Start-up Strategy aimed for company formation to be fully digital and, where possible, completed within 24 hours, and announced a digital funding portal (Federal Government of Germany, 28 July 2022, p. 8).

This was a mixed policy. Grants and advice for SMEs' digital adoption ran, digital hubs linked start-ups with the Mittelstand, and a 24-hour digital formation would cut the cost of entry that, it has been found, holds back the creation of new firms (Klapper et al., 2006). But the formation aim and the portal were announcements, and the online services were still being implemented weeks before the deadline set by law.

## 2.5 HUMAN CAPITAL: EDUCATION AND TRAINING, AND LABOUR

Human capital policy turned on plans to train digital talent and on routes for foreign graduates and specialists; founder routes were plans in Korea and Japan and an announced change in the United Kingdom.

### 2.5.1 KOREA: THE ONE MILLION DIGITAL TALENTS PLAN AND THE VISA PLANS

The plan for digital talent reported to the Cabinet on 22 August aims to train one million people over five years from 2022, 160,000 at beginner, 710,000 at intermediate and 130,000 at advanced level, against about 99,000 trained with public money in 2021, and to at least double school hours in informatics (Ministry of Education, MSIT, Ministry of Employment and Labor, MOTIE, & MSS, 22 August 2022, summary box and background). The Digital Strategy planned an intern visa for students of foreign universities in advanced fields, a negative-list visa for top talent and longer extensions of the technology start-up visa (MSIT, 28 September 2022, strategy, p. 38), and MSS

planned to link its AI education programme to founding and to expand its venture academy from 1,600 trainees in 2022 to 2,400 in 2023 (MSS, 27 December 2022, work plan release, part 1(2)).

This was a mixed policy. Training on this scale, and courses linked to founding, would widen the hiring pool of digital firms. But the targets were a plan and the visa changes, including the longer stay for technology founders, were announcements, so immigrants, who, it has been found, are more likely than similar natives to start companies (Hunt, 2011), gained no new route in the year.

### 2.5.2 JAPAN: THE DIGITAL TALENT TARGET AND THE START-UP VISA REFORM

The Digital Garden City Nation strategy of 23 December aims to train 2.3 million digital personnel over five fiscal years from 2022, the 3.3 million needed less about 1 million existing ICT engineers (Cabinet of Japan, 23 December 2022, strategy, p. 31), and AI Strategy 2022 aims for about 250,000 people a year able to apply data science and AI in their fields (Integrated Innovation Strategy Promotion Council, 22 April 2022, p. 31). The Start-up Development Five-year Plan noted that the start-up visa allowed foreign founders a stay of up to one year, confirmed only by certified local governments, and promised that certified venture capital firms, accelerators and other private bodies would also confirm founders' plans and that the maximum stay would be extended (New Capitalism Realisation Council, 28 November 2022, p. 9).

This was a mixed policy. Letting private bodies confirm founders' plans and lengthening the stay would widen a route for immigrant founders (Hunt, 2011), and the training targets address the shortage of digital personnel. But the visa reform and the targets were plans at the end of the year.

### 2.5.3 CHINA: DIGITAL LITERACY AND SME TRAINING GUIDANCE

The State Council's guiding opinions on digital government of 6 June called for a nationwide system for building digital literacy and skills, with a lifelong digital learning platform and training of officials in digital governance (State Council, 6 June 2022, Part VIII, item (3)). The guide of MIIT's General Office of 3 November advised SMEs to hold regular digital training for managers and front-line staff (General Office of MIIT, 3 November 2022, p. 2). The opinions and the guide named no founders or foreign talent.

This was a poor policy. Literacy for the public and officials and training advice for SMEs carry no budget and do not reach founders, and they open no route for immigrants, whom evidence from the United States shows to start companies at higher rates than similar natives (Hunt, 2011).

### 2.5.4 UNITED KINGDOM: HIGH POTENTIAL INDIVIDUAL AND SCALE-UP ROUTES

The Digital Strategy listed a High Potential Individual route for graduates of top global universities, needing no job offer, from 30 May, and a Scale-up route from 22 August for people with job offers from firms growing by 20 % a year over three years, and it announced that the requirements of the Innovator route would be revised from the autumn, removing the funding requirement where the endorsing body is satisfied (DCMS, 13 June 2022, section 3.6). The AI Action Plan announced funding for up to 2,000 more scholarships for conversion master's courses in AI and data science (DCMS et al., 18 July 2022, action plan, Pillar One).

This was a mixed policy. The new routes widen the hiring pool of digital firms, and immigrants who first entered on student or work visas, it has been found, patent at higher rates than natives (Hunt, 2011). But the change to the Innovator route, the route for founders, was announced without a confirmed start, and the scholarships were an announcement.

### 2.5.5 UNITED STATES: THE STEM DEGREE LIST AND THE CYBER TRAINING ACT

On 21 January the Department of Homeland Security added 22 fields of study, including cloud computing and data science, to the list that qualifies F-1 graduates for a 24-month extension of optional practical training, with effect that day (ICE, 21 January 2022, pp. 3317, 3319). The Small Business Cyber Training Act, enacted on 27 December as Public Law 117-319, requires SBA to set up within 180 days a programme certifying employees of lead Small Business Development Centers in cyber counselling, with reimbursement subject to appropriations (United States Congress, 27 December 2022, sec. 2). The CHIPS and Science Act provided for NSF entrepreneurial fellowships taking ideas from laboratory to market, with funds authorised from fiscal year 2023 (United States Congress, 9 August 2022, sec. 10392(a), (e)).

This was a mixed policy. Longer practical training, in force, keeps foreign graduates in cloud and data fields available to digital firms, and immigrants who first entered as students, it has been found, outperform natives in patenting (Hunt, 2011). But the counselling programme had still to be set up, the fellowships were funded only from fiscal year 2023, and nothing opened a route for founders.

### 2.5.6 FRANCE: THE CYBERSECURITY SKILLS PROJECTS

The projects selected on 27 October under the cybersecurity strategy include a diagnostic of skills needs, starting from the 37,000 cybersecurity jobs that the investment plan expects, and training resources funded under the call on skills and jobs of the future; training more young people and professionals is one of the strategy's five axes (MEFIDS, 27 October 2022, pp. 1, 3). The National Council for Renewal on digital opened a consultation on the digital transition at work, to end in a list of concrete proposals (MEFIDS, 22 November 2022, p. 1), and DeepNum20 offered its start-ups access to the French Tech Visa (French Tech Mission, 27 October 2022, p. 9).

This was a mixed policy. Funded projects address a shortage of cybersecurity skills, and the French Tech Visa reached the start-ups of the cohort. But the council was a consultation, and the year brought no new route for foreign digital talent or founders.

### 2.5.7 GERMANY: THE KEY POINTS ON SKILLED-WORKER IMMIGRATION

On 30 November the Cabinet adopted key points that would lower the EU Blue Card salary thresholds to 1.25 times the average annual gross salary, and 1.0 times for the reduced threshold, ease the salary and German-language rules for IT specialists and introduce an opportunity card for job seekers, with bills to be adopted by the Cabinet in the first quarter of 2023; since June skilled workers had been able to apply online for the Blue Card at some missions abroad (Federal Government of Germany, 30 November 2022, pp. 2, 5–6, 18). The Start-up Strategy sought a fast, digital visa process (Federal Government of Germany, 28 July 2022, p. 7).

This was a mixed policy. Lower thresholds and easier rules for IT specialists would widen digital firms' hiring pool, and online applications had begun. But the reforms were key points, and no route for founders was proposed.

## 2.6 MARKET: MARKET AND INDUSTRY SUPPORT, IPO AND EXIT

Market policy combined action on platform power, public purchasing of IT and digital services, and help for digital exports and trade; none addressed the listing of digital firms.

### 2.6.1 KOREA: GLOBAL-FIRM COLLABORATION AND THE DIGITAL EXPORT PLANS

MSS's budget proposal of 31 August would raise its programme of collaboration with global firms such as Google Play, Nvidia, Microsoft and Siemens from KRW 30.0 billion in 2022 to KRW 40.5 billion in 2023 (MSS, 31 August 2022, part 1). The Digital Strategy planned, from 2023, a digital export pioneer group of large digital firms, SMEs and ventures, and digital representative offices in Silicon Valley, Brussels and Shanghai (MSIT, 28 September 2022, strategy, p. 39), and the FSC planned to help fintech firms expand abroad, starting with South-East Asian countries (FSC, 26 December 2022, part 3).

This was a mixed policy. Programmes with global platform firms give start-ups access to foreign markets and to the platforms on which, it has been argued, many digital firms depend in a relation of unequal power (Cutolo & Kenney, 2021). But the rise was a proposal, and the export group and the offices were plans.

### 2.6.2 JAPAN: THE DIGITAL MARKETPLACE AND THE MOBILE ECOSYSTEM REVIEW

The Digital Agency's priority plan of 7 June set a review, within fiscal year 2022, of IT-system procurement rules to allow fair and quick entry, agile development and cloud, and a study of a digital marketplace to widen start-ups' participation (Digital Agency, 7 June 2022, p. 98); the Start-up Development Five-year Plan provided for the marketplace to be piloted within fiscal year 2023 (New Capitalism Realisation Council, 28 November 2022, p. 15). The New Capitalism plan committed the government to evaluate, within the fiscal year, the platform operators regulated under the Transparency Act and to add digital advertising to the Act (Cabinet of Japan, 7 June 2022, p. 23). On 22 December the secretariat of the Digital Market Competition Council still listed rules for the mobile ecosystem, including an ex-ante framework, as an open question (Cabinet Secretariat of Japan, 22 December 2022, pp. 2, 25).

This was a mixed policy. A marketplace for IT procurement would give start-ups early public buyers, as it has been argued procurement can (Edler & Georghiou, 2007), and the Transparency Act's promised evaluation would apply rules in force to large platforms. But the marketplace and the procurement review were a study and a pilot, and rules for app stores, on which app developers depend, were still under study.

### 2.6.3 CHINA: DATA TRADING VENUES AND CROSS-BORDER E-COMMERCE ZONES

The Data 20 Articles called for measures for data trading venues, whose number is to be strictly controlled, and for data intermediaries to be nurtured (CPC Central Committee & State Council, 2 December 2022, Part III, arts. (9)–(10)). On 14 November the State Council approved 33 more cities and regions as cross-border e-commerce pilot zones and asked them to explore technical standards, processes and supervision for business-to-business trade (State Council, 14 November 2022, items 1, 4). The Government Work Report of 5 March asked large platforms to cut fees and ease small merchants' burden (State Council, 5 March 2022, Part III, section (2)).

This was a mixed policy. Pilot zones and data intermediaries open trade and data markets to small sellers, and lower platform fees ease the costs of firms that, it has been argued, depend on platforms (Cutolo & Kenney, 2021). But the trading rules were still to come, and the fee line was a statement.

### 2.6.4 UNITED KINGDOM: THE DIGITAL MARKETS AND PROCUREMENT BILLS

The Queen's Speech of 10 May announced a draft Digital Markets, Competition and Consumer Bill that would let the Digital Markets Unit designate a small number of very powerful firms in digital activities and set conduct requirements for them (Prime Minister's Office, 10 May 2022, pp. 34–35); the Autumn Statement said that the bill would be brought forward in the third parliamentary session (HM Treasury, 17 November 2022, para. 3.32). The Procurement Bill would open public contracts to new entrants such as small businesses, create a single digital platform on which suppliers register once, and let buyers build demonstrations and prototype testing into procurement (Cabinet Office, 12 May 2022, bullets and sections on doing business and innovation); it was pending at the end of the year. Crown Commercial Service had renewed the AI Marketplace for public buyers of AI (DCMS et al., 18 July 2022, action plan, Pillar Two).

This was a mixed policy. Conduct requirements for the most powerful platforms address the power that, it has been argued, platform owners hold over dependent firms (Cutolo & Kenney, 2021), and the procurement reform and the AI marketplace would let public buyers reach young suppliers (Edler & Georghiou, 2007). But the digital markets bill had not been introduced, and the procurement bill had not been enacted.

### 2.6.5 UNITED STATES: MEMORANDUM M-22-18 AND THE CHIP EXPORT CONTROLS

OMB memorandum M-22-18 of 14 September requires agencies to obtain from producers of third-party software a self-attestation of secure development, within 270 days for critical software and 365 days for all software, and allows them to require a third-party assessment (OMB, 14 September 2022, pp. 3–6). BIS's interim final rule of 13 October controls exports to China of advanced computing chips and some chip-making items, and US persons' support for advanced chip production there, citing supercomputing and AI (BIS, 13 October 2022, p. 62186). Executive Order 14083 of 15 September directed CFIUS to weigh effects on US technological leadership in areas including AI, cybersecurity risks, and access to US persons' sensitive data when it reviews foreign investment (President of the United States, 15 September 2022, secs 2(b), 3(b)–(c)).

This was a mixed policy. Attestation makes software sold to the government more trustworthy, and the controls and the review protect security. But attestation is a fixed cost that weighs most on small software firms selling to agencies, and the controls and the review narrow the markets and investors open to AI and chip firms.

### 2.6.6 FRANCE: THE TRUSTED CLOUD PARTNERSHIP AND INVOICING PLATFORMS

On 26 October the ministers welcomed NUMSPOT, a trusted-cloud offer announced jointly by Docaposte, Dassault Systèmes, Bouygues Telecom and the Banque des Territoires and to be launched during 2023; the economy minister urged administrations and firms to take up sovereign offers, and the cloud strategy aims for French and European champions to emerge (MEFIDS, 26 October 2022, p. 1). The electronic invoicing decree of 7 October sets the conditions for becoming a partner dematerialisation platform and applies the invoicing rules to public procurement contracts (Prime Minister of France, 7 October 2022, Arts. 1–2).

This was a mixed policy. The status of partner platform creates a regulated market for invoicing software firms, and public demand for trusted cloud would give providers early buyers (Edler & Georghiou, 2007). But the cloud measures favoured a partnership of large groups, and the call to buyers was an appeal.

### 2.6.7 GERMANY: THE SECTION 19A DECISIONS AND THE PROCUREMENT PLANS

The Federal Cartel Office decided that Alphabet and its subsidiary Google (announced on 5 January), Meta (4 May) and Amazon (decided on 5 July) are of paramount significance for competition across markets, which places each under the extended abuse control of section 19a of the Competition Act for five years (for Meta, from the date the decision becomes final); Google and Meta declared that they would not appeal (Federal Cartel Office, 5 January 2022, paras 1, 9, 12; Federal Cartel Office, 4 May 2022, paras 1, 9–10; Federal Cartel Office, 6 July 2022, paras 1, 7). On 14 June it opened a proceeding on Apple's rules for tracking by third-party apps, on the suspicion that they favour Apple's own offers (Federal Cartel Office, 14 June 2022, paras 1, 3–4). The Start-up Strategy planned an e-marketplace for innovative procurement at the federal competence centre for innovative procurement, and AI vouchers for SMEs that use AI with a start-up (Federal Government of Germany, 28 July 2022, pp. 13, 15).

The section 19a decisions were a good policy: they put under special abuse control the platform owners on which, it has been argued, many digital firms depend (Cutolo & Kenney, 2021), Amazon among them, whose entry into its sellers' product spaces has been found to discourage the sellers affected (Zhu & Liu, 2018); the Apple proceeding was still open. The procurement plans were a mixed policy: an e-marketplace and AI vouchers would give start-ups first buyers (Edler & Georghiou, 2007), but they were plans.

# 3 ANALYSIS OF THE POLICIES

## 3.1 CONTEXTUAL ANALYSIS: ORIGINS, BACKGROUND AND RATIONALE OF THE POLICIES

Policy in 2022 was made amid changes of government. Korea's new government took over from its predecessor, under which the FSC had issued its plan on cloud rules in April (FSC, 14 April 2022). In the United Kingdom the Autumn Statement of 17 November reversed nearly all the measures of the Growth Plan of September (HM Treasury, 17 November 2022, p. 6). In France a Prime Minister was appointed by decree of 16 May and the government composed on 4 July, with a minister delegate for the digital transition (President of the French Republic, 29 July 2022, preamble and Art. 1). Each government represented the problem of digital entrepreneurship in its own way, and the representation shaped the solutions it chose (Bacchi, 2009; Rochefort & Cobb, 1993) (Table 2).

***Table 2: How the governments represented the problem***

| Country | Problem represented | Presupposition | Left unsaid |
|---|---|---|---|
| Korea | About 99,000 trained with public money in 2021, against a demand of about 738,000 over five years (Ministry of Education et al., 22 August 2022, background) | Training, public data and AI infrastructure will let firms digitalise | Buyers for digital start-ups; start-ups' access to AI models |
| Japan | Analogue rules hold back new technology and start-ups (Digital Agency, 21 December 2022, p. 2) | Removing analogue rules and licensing stablecoins will let digital firms grow | Start-ups' access to computing; when platform rules will come |
| China | SMEs are unwilling, not knowing how, afraid or unable to transform digitally (General Office of MIIT, 25 August 2022, Part I) | State-led data rules, computing hubs and pilots spread digital technology | New ventures and founders, seldom named |
| United Kingdom | Unclear law on AI hurts smaller businesses (DCMS et al., 18 July 2022, policy paper, key challenges) | Existing regulators can govern AI without new law | A successor to the voucher scheme; compute for start-ups |
| United States | Digital-asset firms should face "same business, same risks, same rules" (President of the United States, 9 March 2022, sec. 2(b)) | Executive direction and voluntary frameworks suffice | Support for digital founders; the role of Congress |
| France | The SecNumCloud qualification is demanding and costly (MEFIDS, 22 December 2022, p. 1) | Sovereign cloud and security champions secure the digital economy | Large groups' share of the cloud measures; digital talent |
| Germany | Start-ups and the Mittelstand need better access to data (Federal Government of Germany, 26 January 2022, p. 62) | Strategies, sandboxes and open data will free start-ups to grow | When the announced laws will come; visas for founders |

Source: IFEED analysis after Rochefort and Cobb (1993) and Bacchi (2009) of the sources cited.

Much of the year's money was proposed rather than appropriated. Japan's draft second supplementary budget added JPY 1,327.8 billion of general-account spending for the Cabinet Office (Cabinet Office of Japan, November 2022, p. 2), and Germany's draft budget for 2023 put the digital agenda area at about EUR 1.1 billion (Federal Government of Germany, 5 August 2022, p. 1069). The United Kingdom announced a cut in SME R&D relief but kept larger seed relief (HM Treasury, 17 November 2022, paras 3.31, 5.52), and Japan's outline narrowed its DX tax measure (Cabinet of Japan, 23 December 2022, tax outline, p. 47).

The governments largely shared a rationale: they treated data, cloud and chips as strategic and sought national capacity in them — Korea's vouchers for AI on domestic chips (MSIT, 28 December 2022, para. 3), China's national computing network (NDRC et al., 7 February 2022, item 1), France's aim that French and European cloud champions emerge (MEFIDS, 26 October 2022, p. 1) and the United States' controls on advanced chips (BIS, 13 October 2022, p. 62186). This is the market-creating rationale of mission-oriented innovation policy (Mazzucato, 2016) rather than the correction of particular market failures that entrepreneurship policy has traditionally required (Acs, Åstebro, Audretsch, & Robinson, 2016).

## 3.2 THEORETICAL ANALYSIS: THEORETICAL VALIDITY AND RELEVANCE

Theories of digital entrepreneurship set the first test. They argue that digital ventures arise where infrastructure, data and rules let them reach markets and recombine resources beyond place (Autio, Nambisan, Thomas, & Wright, 2018; Nambisan et al., 2019; Sussan & Acs, 2017). The policies of 2022 passed this test in open data, public digital services and France's DeepNum20, but less so in computing, which in Korea served researchers first and in China came without a route for start-ups (Table 3).

***Table 3: Theoretical validity of the main policies***

| Theory | Claim | Policies consistent with it | Policies in tension with it |
|---|---|---|---|
| Digital entrepreneurial ecosystems (Autio et al., 2018; Sussan & Acs, 2017) | Open data, infrastructure and rules let ventures reach markets and recombine resources | Public data plan (Korea); open data right (Germany); DeepNum20 (France) | AI pilot for researchers (Korea); computing hub without a route for start-ups (China) |
| Platforms (Cutolo & Kenney, 2021; Zhu & Liu, 2018) | Platform owners hold power over dependent firms and enter their markets | Section 19a decisions (Germany); Transparency Act (Japan) | Digital markets bill pending (United Kingdom); app-store rules studied (Japan) |
| Data regulation (Jia et al., 2021) | Uniform data rules cut investment in young, data-related ventures | Volume thresholds for data exports (China) | Deep synthesis duties for all providers (China) |
| Entry regulation (Klapper et al., 2006) | Costly entry regulation holds back the creation of firms | 24-hour digital formation aim (Germany); business login and online subsidies (Japan) | Online services still being built at the deadline (Germany) |
| Grants and R&D tax relief (Agrawal et al., 2020; Howell, 2017) | Early-stage grants raise venture funding; refundable credits raise small firms' R&D | AI vouchers (Korea); security and cloud calls (France); relief for cloud costs (United Kingdom) | SME relief cut (United Kingdom); DX measure narrowed (Japan) |
| Public procurement (Edler & Georghiou, 2007) | Procurement can create early markets for innovation | AI Marketplace (United Kingdom); partner invoicing platforms (France) | Procurement Bill pending (United Kingdom); marketplaces planned (Japan, Germany) |
| Immigrant talent (Hunt, 2011) | Immigrants start firms at higher rates than similar natives | STEM degree list (United States); graduate and scale-up routes (United Kingdom) | Founder visas planned (Korea, Japan); no founder route proposed (Germany) |

Source: IFEED assessment of the policies reviewed in Section 2, against the sources named in the first column.

The evidence on regulation sets the second test. After the European Union's data protection regulation took effect, venture investment fell most for young and data-related ventures (Jia et al., 2021); and where entry regulation is costly, fewer new firms are created (Klapper et al., 2006). China's volume thresholds for data exports and Germany's aim of 24-hour digital formation pass this test; China's deep synthesis duties, which apply whatever a provider's size, are in tension with it.

## 3.3 PRACTICAL ANALYSIS: PERFORMANCE, CONTRIBUTION AND IMPACT

By the end of 2022 much of the year's digital policy had not yet reached firms (Table 4): the sandbox acts, the data and platform bills, the founder visas and most budget lines were plans, bills or drafts. Governments reported mostly outputs: 327 approvals under Korea's industrial convergence sandbox, whose 173 firms in business MOTIE credited with KRW 163.1 billion of sales and 866 jobs (MOTIE, 20 December 2022, summary); 306 consultations at Japan's FinTech Support Desk (FSA, August 2022, p. 174); nearly 150,000 France Num support actions (DGE, 2022, p. 9); and more than 2,000 grants under Germany's Digital Now by the end of 2021, beside an evaluation in which nearly 94 % of SMEs surveyed said they would not have digitalised in the same way without go-digital (Federal Government of Germany, 6 December 2022, pp. 4–5). The United Kingdom reported low take-up of its voucher scheme: fewer than 1,000 vouchers redeemed against a design for 100,000 SMEs (BEIS, 15 December 2022, para. 3). Performance measures serve, among other purposes, to evaluate, to control, to celebrate or to promote (Behn, 2003); those of 2022 served mostly to celebrate and to promote.

Table 5 summarises the assessments of Section 2. A policy is rated good where every measure judged was in force, launched or enacted and reached the firms it was meant for; mixed where such measures sat beside pending or announced ones or reached firms other than founders, or where only pending measures aimed at digital firms were set out; and poor where nothing in force reached digital firms or founders and what was set out was aimed elsewhere.

***Table 4: Status of the main measures at the end of 2022***

| Country | In force or launched in 2022 | Adopted, in force from 2023 | Pending or announced |
|---|---|---|---|
| Korea | Industrial convergence sandbox; hyperscale AI pilot; AI Voucher call; Digital Asset Task Force | Data Industry Act decree; Smart Manufacturing Act (promulgation due) | AI and Digital Society Basic Acts; sandbox act; cloud rules; fintech fund top-up; founder funding; visa changes; public data plan; export plans; budget proposal |
| Japan | FinTech Support Desk; fintech conference | Payment Services Act amendment (in force within a year) | Transparency Act evaluation; G-Biz ID and J-Grants upgrades; payment orders and supplementary budget (drafts); tax outline; bundle bill; visa reform; digital marketplace; platform rules; talent target |
| China | Outbound data measures; SME fiscal pilot; service festival; computing hub; e-commerce pilot zones | Deep synthesis provisions; industry data security measures | Rules under the Data 20 Articles; computing vouchers (encouraged); digital literacy system |

| Country | In force or launched in 2022 | Adopted, in force from 2023 | Pending or announced |
|---|---|---|---|
| United Kingdom | High Potential Individual and Scale-up routes; AI Marketplace | Closure of Help to Grow: Digital | Product Security Act (awaiting regulations); Made Smarter extension; AI white paper; Data Reform, Digital Markets and Procurement Bills; cyber and cryptoasset rules; R&D relief changes; Innovator route |
| United States | Executive Orders 14067 and 14083; AI Bill of Rights Blueprint; M-22-18; chip controls; STEM degree list; SBA pilot; CHIPS and Science Act | NSF fellowships (funded from fiscal year 2023) | CIRCIA (rule to follow); Cyber Training Act (programme due); NIST framework (draft); payments framework; NAIRR |
| France | Cybersecurity and cloud calls; SecNumCloud scheme; DeepNum20; Next40/120; France Num; 5G platforms; council on digital | Platform audit (from October 2023); invoicing duty (phased in later) | Trusted cloud offer; follow-up of EU digital rules |
| Germany | Section 19a decisions; Apple proceeding; Digital Now; go-digital; founding competition; online Blue Card applications | — | Real-World Laboratories Act; open data right; draft budget lines; Future Financing Act; immigration key points; 24-hour formation; e-marketplace; AI vouchers; Start-up Summit |

Source: IFEED compilation from the sources cited in Section 2.

***Table 5: Assessment of the policies by domain and country***

| Domain | Korea | Japan | China | United Kingdom | United States | France | Germany |
|---|---|---|---|---|---|---|---|
| Regulation and Law | Mixed | Mixed | Mixed | Mixed | Mixed | Mixed | Mixed |
| Finance | Mixed | Mixed | Mixed | Mixed | Mixed | Mixed | Mixed |
| Culture | Mixed | Mixed | Mixed | Mixed | Poor | Mixed | Mixed |
| Support | Mixed | Mixed | Mixed | Mixed | Mixed | Good | Mixed |
| Human Capital | Mixed | Mixed | Poor | Mixed | Mixed | Mixed | Mixed |
| Market | Mixed | Mixed | Mixed | Mixed | Mixed | Mixed | Mixed |

Source: IFEED assessment of the policies reviewed in Section 2. Where a subsection assessed its policies differently, Mixed is shown.

# 4 POLICY LIMITATIONS

## 4.1 POLICY DEBATES, CONTESTED POINTS AND ISSUES

The policies of 2022 raised six points of contention, set out in Table 6. The first two concerned rules: how far data duties should be scaled to the size of firms, as China's thresholds for data exports and the United Kingdom's exemption of small and micro businesses from cyber rules do (CAC, 7 July 2022, Art. 4; DCMS, 30 November 2022, section 4.1), and whether AI should be governed by principles and voluntary frameworks or by binding rules such as China's deep synthesis provisions.

The other four concerned money, markets and people: how crypto-assets and stablecoins should be licensed; whether platform power should be checked case by case, as Germany's Federal Cartel Office did, or by rules set in advance, as the United Kingdom's draft bill and Japan's review proposed; whether subsidies for SMEs' digital adoption give value for money, which the United Kingdom questioned when it closed its voucher scheme while extending Made Smarter Adoption; and whether migration control should yield to the supply of digital talent and founders.

***Table 6: Debates and contested points in the policies of 2022***

| Issue | What the governments set out | Point at issue |
|---|---|---|
| Data duties | Volume thresholds for data exports (China), small-firm exemption kept (United Kingdom), data quality certification (Korea), open data right announced (Germany) (CAC, 7 July 2022; DCMS, 30 November 2022; Federal Government of Germany, 28 July 2022; MSIT, 27 December 2022) | Protection of data and security against the fixed costs of young, data-intensive firms |
| AI rules | Principles (Japan, United Kingdom), non-binding blueprint and draft framework (United States), deep synthesis rules (China), AI basic act planned (Korea) (CAC et al., 25 November 2022; DCMS et al., 18 July 2022; Integrated Innovation Strategy Promotion Council, 22 April 2022; MSIT, 28 September 2022; OSTP, October 2022) | Legal certainty against compliance costs that fall most on young firms |
| Crypto-assets and payments | Stablecoin law (Japan), executive order and reports (United States), regime planned (United Kingdom), law under study (Korea) (FSA, 4 March 2022; FSC, 17 August 2022; HM Treasury, 4 April 2022; President of the United States, 9 March 2022) | Licensed innovation against consumer and investor protection |
| Platform power | Section 19a decisions (Germany), draft bill (United Kingdom), ex-ante rules studied (Japan), fee line (China) (Cabinet Secretariat of Japan, 22 December 2022; Federal Cartel Office, 5 January 2022, 4 May 2022, 6 July 2022; Prime Minister's Office, 10 May 2022; State Council, 5 March 2022) | Enforcement case by case against rules set in advance |

| Issue | What the governments set out | Point at issue |
|---|---|---|
| SME digital adoption | Fiscal pilot (China), grants and advice (Germany), France Num (France), AI vouchers (Korea), voucher scheme closed and Made Smarter extended (United Kingdom) (BEIS, 15 December 2022; DGE, 2022; Federal Government of Germany, 6 December 2022; General Offices of MIIT & MOF, 15 August 2022; HM Treasury, 17 November 2022; MSIT, 28 December 2022) | Subsidies for adoption against value for money |
| Talent | Practical training widened (United States), graduate and scale-up routes (United Kingdom), founder visas planned (Korea, Japan), key points for specialists (Germany) (DCMS, 13 June 2022; Federal Government of Germany, 30 November 2022; MSIT, 28 September 2022; New Capitalism Realisation Council, 28 November 2022; ICE, 21 January 2022) | Migration control against the supply of digital talent and founders |

Source: IFEED analysis of the sources cited.

## 4.2 THEORETICAL LIMITATIONS OF THE POLICIES

The first limitation is that the policies treated digital technology as something to be adopted rather than as a source of new ventures. Digital affordances, it has been argued, let new firms find and pursue opportunities beyond place (Autio et al., 2018; Nambisan, 2017), but the year's instruments for support and culture — pilots, guides, festivals, vouchers and advice for SMEs — aimed mostly at users of digital tools, and few reached founders. It has been argued that the weakest condition limits the system (Ács, Autio, & Szerb, 2014), yet the governments assumed, without showing it, that adoption by established firms was the binding condition.

The second is selection. Korea aimed to raise its digital unicorns from 23 to 100 within five years (MSIT, 28 September 2022, strategy, p. 38) and planned to select leading fintech firms, China supported "little giants" and unicorns in AI scenarios (MOST et al., 29 July 2022, item 8), and France's DeepNum20 picked start-ups by jury. High-growth firms, it has been argued, vary too much in sector, age, size and origin for support to be targeted on particular technologies or types of firm (Mason & Brown, 2013).

The third is the uniformity of duties. China's deep synthesis provisions would apply to small firms as to large ones, although uniform data rules have been found to fall hardest on new, data-related ventures (Jia et al., 2021), and the US attestation for software sold to agencies is a fixed cost that weighs most on small sellers; and in the United Kingdom the cut in SME R&D relief will fall on the small firms whose R&D refundable credits have been found to raise (Agrawal et al., 2020). The instrument that scales duties to risk, the sandbox, worked in Korea but was a plan in Germany and the United Kingdom.

The fourth is the weakness of evaluation. Entrepreneurship policy has long been criticised for counting the number of firms rather than their quality and for lacking evaluation (Acs et al., 2016), and mission-oriented policy, it has been argued, needs organisations that learn through discovery and new ways of evaluating it (Mazzucato, 2016). The figures of 2022 repeat the counting: the effects reported were claimed by ministries or surveyed users, and only the United Kingdom explained a closure by its results.

# 5 POLICY IMPLICATIONS

First, data and computing should reach start-ups by rule. Korea should open its hyperscale AI pilot to SMEs and start-ups as it planned to consider, Germany should table its right to open data, China should fund the computing vouchers that it encouraged localities to offer, and Japan and the United Kingdom should state how start-ups will reach public computing.

Second, digital duties should be scaled to the size of firms. China's thresholds for data exports and the United Kingdom's exemption of small and micro businesses show how; China's deep synthesis duties and the US attestation for software sold to agencies should spare or phase in small firms, and Korea and Germany should enact their sandbox laws so that young firms can test products under exemption.

Third, platform rules should be set in law. The United Kingdom should introduce its digital markets bill and Japan should decide on rules for app stores; Germany's section 19a decisions show that a competition authority can act against platform owners once the law allows it.

Fourth, governments should act as paying first customers. Japan should pilot its digital marketplace, Germany should open its e-marketplace for innovative procurement, the United Kingdom should enact its procurement reform, and governments should publish every year the share of their digital purchases made from young firms.

Finally, talent rules and evaluation should serve digital founders. Korea and Japan should enact their visa changes for founders, the United Kingdom should confirm the start of its revised Innovator route, and Germany should add a route for founders to its immigration reform. Governments should measure and publish the effects of their digital programmes, as the United Kingdom began to do when it reported the take-up of its closed voucher scheme and evaluated its Global Talent visa process (Home Office, 30 May 2022), and the United Kingdom should keep relief for small firms' R&D while extending it to data and cloud costs.

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